

W3 Levels of Abstraction

Thursday, October 8, 2026

READINGS Marr, *Vision* (1982), ch. 1, excerpts; Newell, “The Knowledge Level” (1982), excerpts

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1 Levels of explanation, the very idea

The doctrine of levels

- **Levels.** Descriptions of complex systems can be divided into levels. Explanations which identify similar patterns group together into levels.
- **Levels. Ordering.** There is some kind of asymmetrical dependence ordering on levels of explanation.
- **Abstraction and realization.** Upper levels are abstractions of lower levels. Lower levels are realizations of upper levels.
- **Real patterns.** Upper levels capture real patterns that emerge from lower levels.
- **Levels in nature.** The natural sciences as a whole are organized into levels of abstraction, also known as levels of explanation.
- **Every level is necessary.** A complete explanation of reality requires the contributions of all levels, not just the bottom level of physics.
- **Every level is autonomous.** Levels of explanation can be studied at least in part without knowledge of the levels above and below.

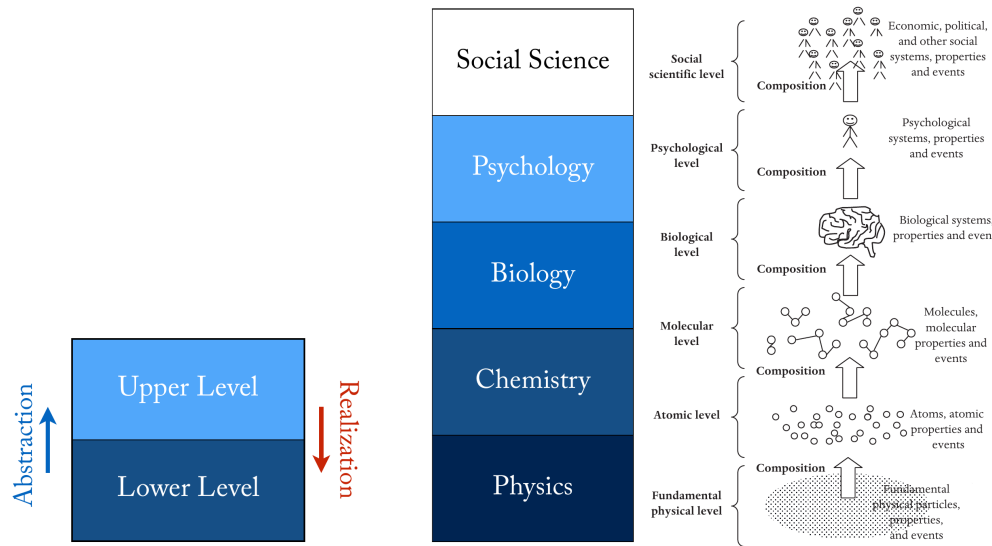


Figure 1. Levels in physical reality.

2 Marr's Levels

Analyzing complex systems

- **Complex systems.** "Almost never can a complex system of any kind be understood as a simple extrapolation from the properties of its elementary components." (p. 19)
- **Different kinds of explanation.** "If one hopes to achieve a full understanding of a system as complicated as a nervous system, a developing embryo, a set of metabolic pathways, a bottle of gas, or even a large computer program, then one must be prepared to contemplate different kinds of explanation at different levels of description that are linked, at least in principle, into a cohesive whole, even if linking the levels in complete detail is impractical." (p. 20)
- **The brain.** The brain itself is a complex system which must be studied at many levels simultaneously.

Marr's three levels

Computational theory	Representation and algorithm	Hardware implementation
What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?	How can this computational theory be implemented? In particular, what is the representation for the input and output, and what is the algorithm for the transformation?	How can the representation and algorithm be realized physically?

Figure 2. “The three levels at which any machine carrying out an information-processing task must be understood.”

- **Level 1: Computation.** Explanations at this level have two halves: (a) a description of *what* (function) is being computed; (b) an account of *why* that function is being computed.
- **Level 2: Algorithm.** Explanations at this stage specify both the system of representation and the algorithm being used. We should expect a wealth of sub-levels within the algorithm level. Algorithms are typically describable at multiple distinct levels of abstraction.
- **Level 3: Implementation.** Again, we should expect a wealth of sub-levels within the implementation level.
- **Independence of levels.** The three levels are only “loosely” related; they can and must be studied both independently and together.

Example: the humble cash register

- **Computational level.** “The most abstract is the level of *what* the device does and *why*. What it does is arithmetic, so our first task is to master the theory of addition.” (p. 22)
- **Addition.** “Addition is a mapping, usually denoted by $+$, from pairs of numbers into single numbers; for example, $+$ maps the pair $(3, 4)$ to 7, and I shall write this in the form $(3 + 4) \rightarrow 7$.” (p. 22)
- **Abstract properties,** “Addition has a number of abstract properties, however. It is commutative: both $(3 + 4)$ and $(4 + 3)$ are equal to 7; and associative: the sum of $3 + (4 + 5)$ is the same as the sum of $(3 + 4) + 5$. Then there is the unique distinguished element, zero, the adding of which has no effect: $(4 + 0) \rightarrow 4$. Also, for every number

there is a unique ‘inverse,’ written (-4) in the case of 4, which when added to the number gives zero: $[4 + (-4)] \rightarrow 0$.” (p. 22)

- **What is computed.** “Notice that these properties are part of the fundamental theory of addition. They are true no matter how the numbers are written—whether in binary, Arabic, or Roman representation—and no matter how the addition is executed. Thus part of this first level is something that might be characterized as what is being computed.” (p. 22)
- **Why.** “The other half of this level of explanation has to do with the question of why the cash register performs addition and not, for instance, multiplication when combining the prices of the purchased items to arrive at a final bill.” (p. 22)
 - (1) If you buy nothing, it should cost you nothing; and buying nothing and something should cost the same as buying just the something. (The rules for zero.)
 - (2) The order in which goods are presented to the cashier should not affect the total. (Commutativity.)
 - (3) Arranging the goods into two piles and paying for each pile separately should not affect the total amount you pay. (Associativity; the basic operation for combining prices.)
 - (4) If you buy an item and then return it for a refund, your total expenditure should be zero. (Inverses.)
- **Computational theory.** “This whole argument is what I call the *computational theory* of the cash register. Its important features are (1) that it contains separate arguments about what is computed and why and (2) that the resulting operation is defined uniquely by the constraints it has to satisfy.” (p. 23)
 - **Vision.** “In the theory of visual processes, the underlying task is to reliably derive properties of the world from images of it; the business of isolating constraints that are both powerful enough to allow a process to be defined and generally true of the world is a central theme of our inquiry.” (p. 23)
- **Algorithmic level.** “The second level of the analysis of a process, therefore, involves choosing two things: (1) a *representation* for the input and for the output of the process and (2) an *algorithm* by which the transformation may actually be accomplished.” (p. 23)
 - **What, why, how.** “If the first of our levels specifies what and why, this second level specifies how. For addition, we might choose Arabic numerals for the

representations, and for the algorithm we could follow the usual rules about adding the least significant digits first and ‘carrying’ if the sum exceeds 9.” (p. 23)

- **Hardware level.** “This brings us to the third level, that of the device in which the process is to be realized physically. The important point here is that, once again, the same algorithm may be implemented in quite different technologies.” (p. 24)

Example: the +1 machine

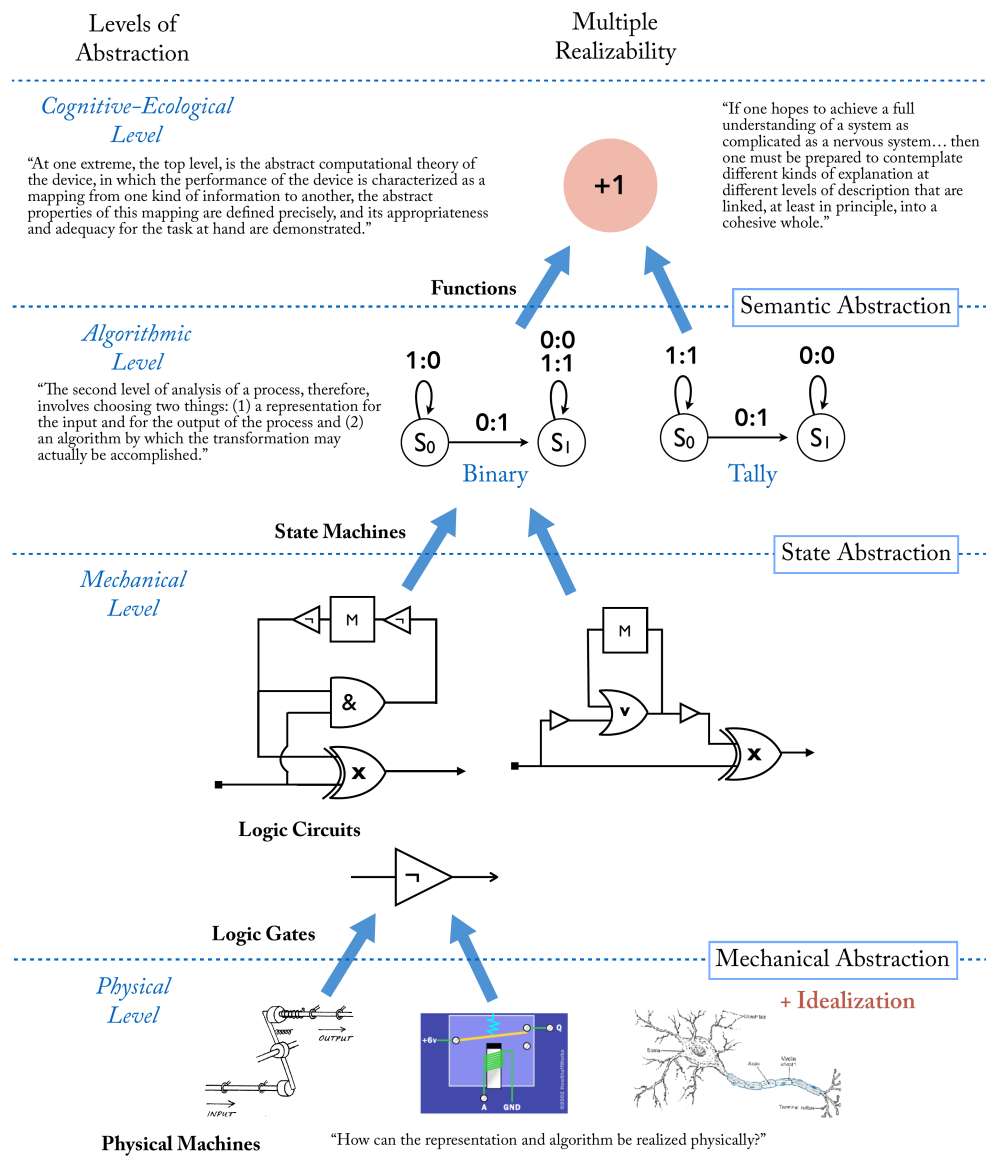


Figure 3. Levels of abstraction

- **Phenomena at different levels.** “In attempts to relate psychophysical problems to physiology, too often there is confusion about the level at which problems should be addressed. For instance, some are related mainly to the physical mechanisms

of vision—such as afterimages (for example, the one you see after staring at a light bulb) or such as the fact that any color can be matched by a suitable mixture of the three primaries (a consequence principally of the fact that we humans have three types of cones).” (p. 25)

Example: The Necker cube

- **The Necker cube.** “On the other hand, the ambiguity of the Necker cube (Figure 1–5) seems to demand a different kind of explanation. To be sure, part of the explanation of its perceptual reversal must have to do with a bistable neural network (that is, one with two distinct stable states) somewhere inside the brain, but few would feel satisfied by an account that failed to mention the existence of two different but perfectly plausible three-dimensional interpretations of this two-dimensional image.” (pp. 25–26)

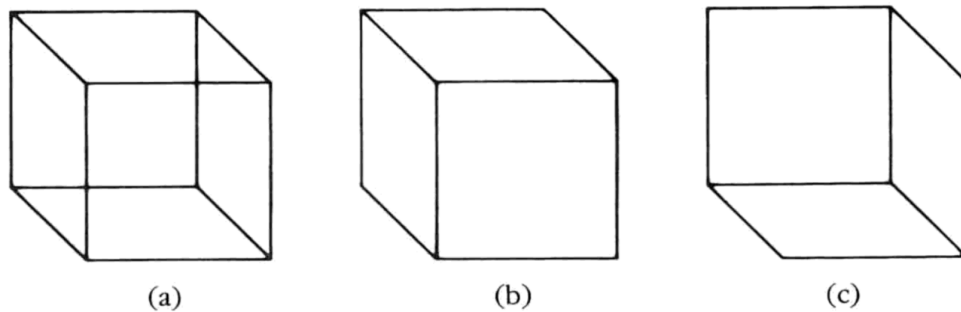


Figure 4. Marr’s Figure 1–5 (p. 26): “The so-called Necker illusion, named after L. A. Necker, the Swiss naturalist who developed it in 1832. The essence of the matter is that the two-dimensional representation (a) has collapsed the depth out of a cube and that a certain aspect of human vision is to recover this missing third dimension. The depth of the cube can indeed be perceived, but two interpretations are possible, (b) and (c). A person’s perception characteristically flips from one to the other.”

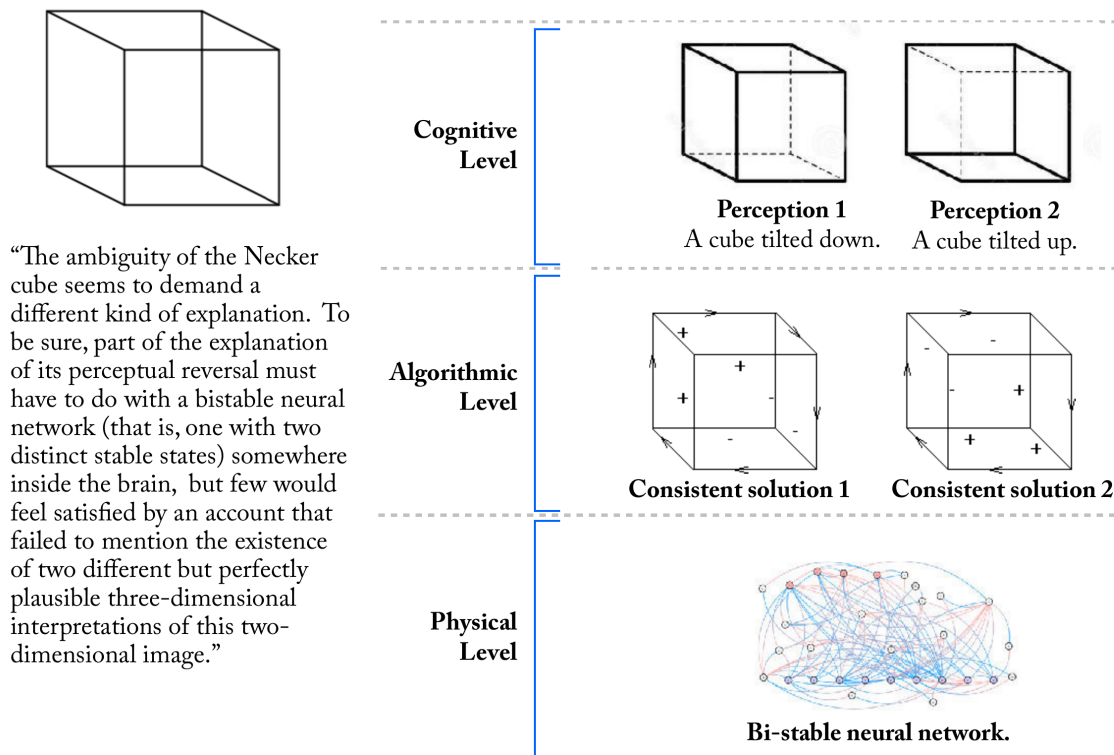


Figure 5. The Necker cube at three levels.

Importance of computational theory

- **The top level.** “Although algorithms and mechanisms are empirically more accessible, it is the top level, the level of computational theory, which is critically important from an information-processing point of view. The reason for this is that the nature of the computations that underlie perception depends more upon the computational problems that have to be solved than upon the particular hardware in which their solutions are implemented. To phrase the matter another way, an algorithm is likely to be understood more readily by understanding the nature of the problem being solved than by examining the mechanism (and the hardware) in which it is embodied.” (p. 27)
- **Feathers.** “In a similar vein, trying to understand perception by studying only neurons is like trying to understand bird flight by studying only feathers: It just cannot be done. In order to understand bird flight, we have to understand aerodynamics; only then do the structure of feathers and the different shapes of birds’ wings make sense.” (p. 27)
- **The cognitive-ecological level.** Marr called the top level of his hierarchy the “computational” level, with an eye towards *what* is computed. But this is confusing to us, since “computation” evokes algorithms and circuits. Instead, I’ll call the

top level the cognitive-ecological level. It is the level where the cognitive task is specified and where the ecological rationale for performing that task is explained.

- **Chomsky's theory as computational theory.** "Failure to recognize this theoretical distinction between what and how also greatly hampered communication between the fields of artificial intelligence and linguistics. Chomsky's (1965) theory of transformational grammar is a true computational theory in the sense defined earlier. It is concerned solely with specifying what the syntactic decomposition of an English sentence should be, and not at all with how that decomposition should be achieved." (p. 28)
- **Competence and performance.** "Chomsky himself was very clear about this—it is roughly his distinction between competence and performance, though his idea of performance did include other factors, like stopping in midutterance—but the fact that his theory was defined by transformations, which look like computations, seems to have confused many people." (p. 28)
- **Theory and implementation.** "The explanation is simply that finding algorithms by which Chomsky's theory may be implemented is a completely different endeavor from formulating the theory itself. In our terms, it is a study at a different level, and both tasks have to be done." (p. 28)

3 Marr: the function of vision

The purpose of vision

- **Many purposes.** "A pigeon uses vision to help it navigate, fly, and seek out food. Many types of jumping spider use vision to tell the difference between a potential meal and a potential mate." (p. 32)
- **The ur-purpose:** image into shape
 - "And second, vision alone can deliver an internal description of the shape of a viewed object, even when the object was not recognized in the conventional sense of understanding its use and purpose." (p. 35)
 - "The second important thing, I thought, was that Elizabeth Warrington had put her finger on what was somehow the quintessential fact of human vision—that it tells about shape and space and spatial arrangement. Here lay a way to formulate its purpose—building a description of the shapes and positions of things from images." (p. 36)

- “But these things [illumination, reflectance, motion] seemed secondary; they could be hung off a theory in which the main job of vision was to derive a representation of shape.” (p. 36)

To the desirable via the possible

Name	Purpose	Primitives
Image(s)	Represents intensity.	Intensity value at each point in the image
Primal sketch	Makes explicit important information about the two-dimensional image, primarily the intensity changes there and their geometrical distribution and organization.	Zero-crossings Blobs Terminations and discontinuities Edge segments Virtual lines Groups Curvilinear organization Boundaries
2½-D sketch	Makes explicit the orientation and rough depth of the visible surfaces, and contours of discontinuities in these quantities in a viewer-centered coordinate frame.	Local surface orientation (the “needles” primitives) Distance from viewer Discontinuities in depth Discontinuities in surface orientation
3-D model representation	Describes shapes and their spatial organization in an object-centered coordinate frame, using a modular hierarchical representation that includes volumetric primitives (i.e., primitives that represent the volume of space that a shape occupies) as well as surface primitives.	3-D models arranged hierarchically, each one based on a spatial configuration of a few sticks or axes, to which volumetric or surface shape primitives are attached

Figure 6. Marr’s Table 1–1 (p. 37): “Representational framework for deriving shape information from images.”

Marr’s theory of vision

- **The purpose of vision.** 3D → 2D → 3D.

- **The world: 3D $\rightarrow$ 2D.** The world projects a 3D scene onto a 2D array of light. Information is lost in this projection.
- **The visual system: 2D $\rightarrow$ 3D.** The purpose of vision is to solve the inverse projection problem; i.e., to reconstruct the distal cause from the proximal effect. (Compare to Helmholtz.)
- **Heuristic expectations.** Because information is lost at the world stage, the inverse projection problem cannot be solved in general. It can only be solved approximately, using heuristic expectations about the world which are encoded in the perceptual machinery.
- **Local modularity.** The visual system itself is divided into interconnected but largely modular computational processes. The results of such modular processing can be isolated.
- **Layers of processing.** According to Marr the passage from 2D stimulus to 3D description can be understood as a passage from a 2D viewer-centered description to a 3D object-centered description, realized by a series of intermediate representations.



Figure 1–1. A random-dot stereogram of the type used extensively by Bela Julesz. The left and right images are identical except for a central square region that is displaced slightly in one image. When fused binocularly, the images yield the impression of the central square floating in front of the background.

Figure 7. Random-dot stereogram.

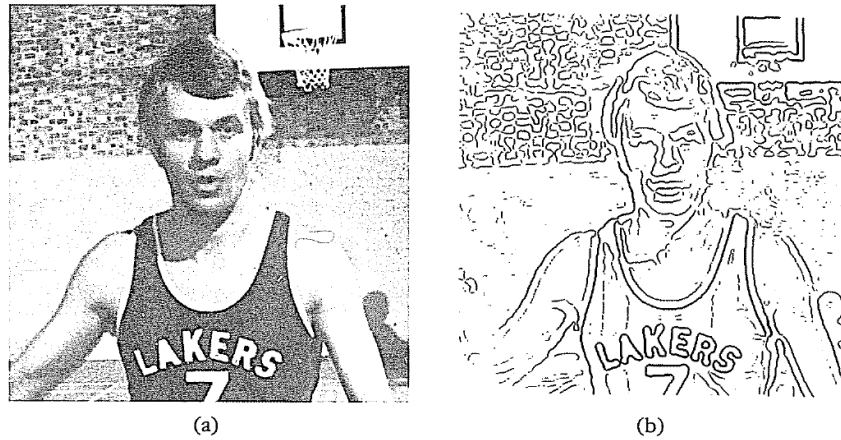


Figure 2-15. Another example of zero-crossings; here, the intensity of the lines has been made to vary with the slope of the zero-crossing, so that it is easier to see which lines correspond to the greater contrast. (Courtesy BBC Horizon.)

Figure 8. Edge-detection.

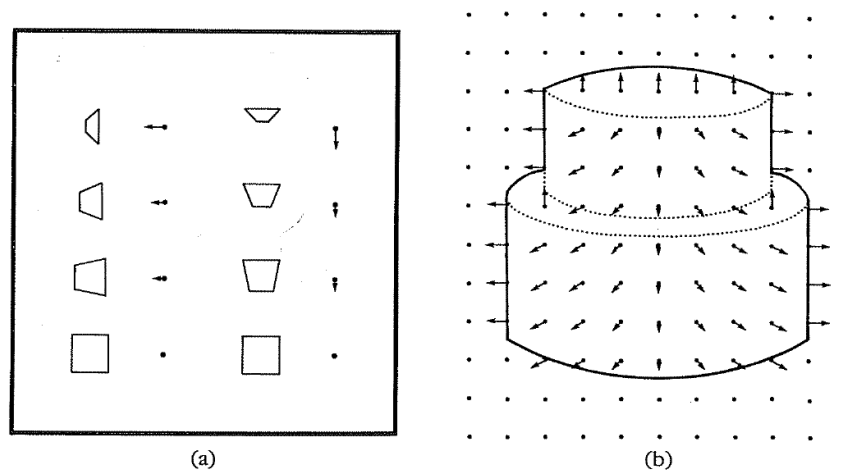


Figure 3-12. Illustration of the $2\frac{1}{2}$ -dimensional sketch. In (a), the perspective views of small squares placed at various orientations to the viewer are shown. The dots with arrows symbolically represent the orientations of such surfaces. In (b), this symbolic representation is used to show the surface orientations of two cylindrical surfaces in front of a background orthogonal to the viewer. The full $2\frac{1}{2}$ -dimensional sketch would include rough distances to the surfaces as well as their orientations; contours where surface orientations change sharply, which are shown dotted; and contours where depth is discontinuous (subjective contours), which are shown with full lines. See Chapter 4 for more details. (D. Marr and H. K. Nishihara, 1978.)

Figure 9. The $2\frac{1}{2}$ -D sketch.

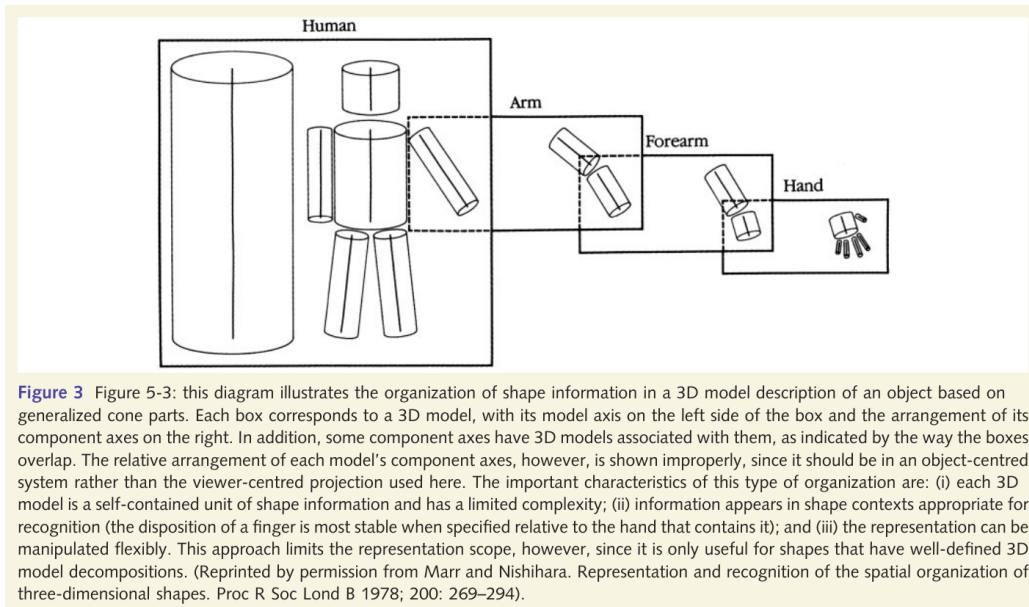
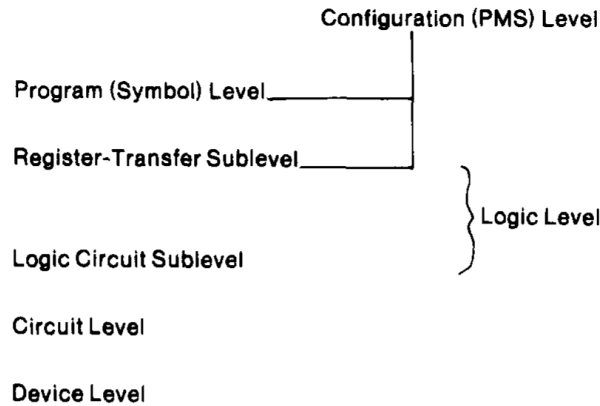


Figure 10. The 3-D model representation

4 Newell: the knowledge level

Computer system levels

- **What is a level?** “A level consists of a medium that is to be processed, components that provide primitive processing, laws of composition that permit components to be assembled into systems, and laws of behavior that determine how system behavior depends on the component behavior and the structure of the system.” (p. 95)
- **Autonomy.** “Each level is defined in two ways. First, it can be defined autonomously, without reference to any other level.” (p. 95)
- **Reduction.** “Second, each level can be reduced to the level below.” (p. 95)
- **Variety.** “However, it is noteworthy how radically the levels differ. The medium changes from electrons and magnetic domains at the device level, to current and voltage at the circuit level, to bits at the logic level (either single bits at the logic circuit level or bit vectors at the register-transfer level), to symbolic expressions at the symbol level, to amounts of data (measured in data bits) at the configuration level.” (pp. 95–96)



(a) Newell's Fig. 2 (p. 96): "Computer system levels."

Aspects	Register-transfer level	Symbol level
Systems	Digital systems	Computers
Medium	Bit vectors	Symbols, expressions
Components	Registers	Memories
	Functional units	Operations
Composition laws	Transfer path	Designation, association
Behavior laws	Logical operations	Sequential interpretation

(b) Newell's Table 1 (p. 96): "Defining aspects of a computer system level."

Figure 11. Newell's hierarchy of computer system levels.

- **Basic features.** "Despite this variety, all levels share some common features." (p. 96)
 - **Point 1.** "Specification of a system at a level always determines completely a definite behavior for the system at that level (given initial and boundary conditions)." (p. 96)
 - **Point 2.** "The behavior of the total system results from the local effects of each component of the system processing the medium at its inputs to produce its outputs." (p. 96)
 - **Point 3.** "The immense variety of behavior is obtained by system structure, i.e., by the variety of ways of assembling a small number of component types (though perhaps a large number of instances of each type)." (p. 96)
 - **Point 4.** "The medium is realized by state-like properties of matter, which remain passive until changed by the components." (p. 97)
- **Realism.** "To sum up, computer system levels are a reflection of the nature of the physical world. They are not just a point of view that exists solely in the eye

of the beholder. This reality comes from computer system levels being genuine specializations, rather than being just abstractions that can be applied uniformly.” (p. 98)

The knowledge level

- **The proposal.** “I now propose that there does exist yet another system level, which I will call the knowledge level. It is a true systems level, in the sense we have just reviewed.” (p. 98)
- **The agent.** “The system at the knowledge level is the agent. The components at the knowledge level are goals, actions, and bodies. Thus, an agent is composed of a set of actions, a set of goals and a body. The medium at the knowledge level is knowledge (as might be suspected). Thus, the agent processes its knowledge to determine the actions to take. Finally, the behavior law is the principle of rationality: Actions are selected to attain the agent’s goals.” (p. 98)
- **Above the symbolic level.** “The knowledge level sits in the hierarchy of systems levels immediately above the symbol level, as Fig. 3 shows. Its components (actions, goals, body) and its medium (knowledge) can be defined in terms of systems at the symbol level, just as any level can be defined by systems at the level one below.” (pp. 98–99)

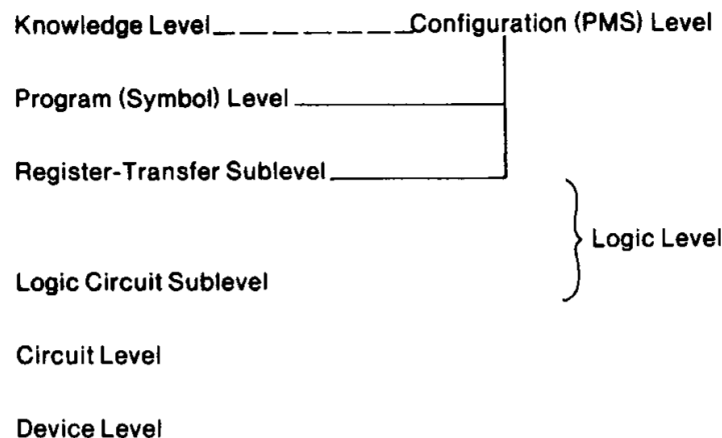


Figure 12. Newell’s Fig. 3 (p. 99): “New version of computer system levels.”

- **The Knowledge Level Hypothesis.** “There exists a distinct computer systems level, lying immediately above the symbol level, which is characterized by knowledge as the medium and the principle of rationality as the law of behavior.” (p. 99)

- **Knowledge and rationality.** “Knowledge is intimately linked with rationality. Systems of which rationality can be posited can be said to have knowledge. It is unclear in what sense other systems can be said to have knowledge.” (p. 100)

The structure of the knowledge level

- **The body.** “First, the agent has some physical body with which it can act in the environment (and be acted upon). We talk about the body as if it consists of a set of actions, but that is only for simplicity. It can be an arbitrary physical system with arbitrary modes of interaction with its environment.” (pp. 100–101)
- **Knowledge.** “Second, the agent has a body of knowledge. This body is like a memory. Whatever the agent knows at some time, it continues to know.” (p. 101)
- **Goals.** “Third, and finally, the agent has a set of goals. A goal is a body of knowledge of a state of affairs in the environment. Goals are structurally distinguished from the main body of knowledge. This permits them to enter into the behavior of the agent in a distinct way, namely, that which the organism strives to realize. But, except for this distinction, goal components are structurally identical to bodies of knowledge.” (p. 101)
- **What behavior depends on.** “The behavior is to depend only what the agent knows, what it wants and what means it has for interacting physically with the environment.” (p. 101)
- **Absence of structure.**
 - **No structural constraints.** “There are no structural constraints to the knowledge in a body, either in capacity (i.e., the amount of knowledge) or in how the knowledge is held in the body. Indeed, there is no notion of how knowledge is held (encoding is a notion at the symbol level, not knowledge level).” (p. 101)
 - **Essentially structure-less.** “This complete absence of significant structure in the agent is the first surprise, running counter to the common feature at all levels that variety of behavior is realized by variety of system structure (Section 3.1, Point 3). This is not fortuitous, but is an essential feature of the knowledge level. The focus for determining the behavior of the system rests with the knowledge, i.e., with the content of what is known. The internal structure of the system is defined exactly so that nothing need be known about it to predict the agent’s behavior.” (p. 101)
 - **Compare: possible-world analysis of propositions.**

The principle of rationality

- **Principle of rationality.** “If an agent has knowledge that one of its actions will lead to one of its goals, then the agent will select that action.” (p. 102)
- **No mechanism.** “This principle asserts a connection between knowledge and goals, on the one hand, and the selection of actions on the other, without specification of any mechanism through which this connection is made. It connects all the components of the agent together directly.” (p. 102)
- **Not sufficient to determine behavior.** “However, the principle is not sufficient to determine the behavior in many situations. Some of these situations can be covered by adding auxiliary principles. Thus, we can think of an extended principle of rationality, building out from the central or main principle, given above.” (p. 102)
- **The lady or the tiger.** “Our knowledge-level model of the princess, even if it were to include her complete knowledge, would not tell us which she chose. But the failure of determinacy is not the model’s.” (p. 104)
 - **Radical incompleteness.** “Yet, radical incompleteness characterizes the knowledge level.” (pp. 104–105)

The nature of knowledge

- **Knowledge defined.** “Whatever can be ascribed to an agent, such that its behavior can be computed according to the principle of rationality.” (p. 105)
 - **Knowledge as...**
 - * Knowledge
 - * Belief
 - * Committal mental states
 - * Committal dispositions.
- **Relation to symbols and processes.** “This still leaves open the requirement for a physical structure for knowledge that can fill the functional role. In fact, that key role is never filled directly. Instead, it is filled only indirectly and approximately by symbol systems at the next lower level. These are total systems, not just symbol structures. Thus, knowledge, though a medium, is embodied in no medium-like passive physical structure.” (p. 105)
 - **No physical structure.** “This failure to have the medium at the knowledge level be a state-like physical structure is the fourth surprise, running counter to the common feature at all levels of a passive medium (Section 3.1, Point 4). Again, it is an essential feature of the knowledge level, giving knowledge its special abstract and competence-like character.” (p. 105)

Why is knowledge irreducible to symbols?

- **The question.** “Even granting the scheme of Fig. 4 to be possible, why cannot there simply be some physical structure that corresponds to knowledge?” (p. 107)
- **Knowledge is not finite.** “The answer in a nutshell is that knowledge of the world cannot be captured in a finite structure. The world is too rich and agents have too great a capability for responding. Knowledge is about the world. Insofar as an agent can select actions based on some truth about the world, any candidate structure for knowledge must contain that truth. Thus knowledge as a structure must contain at least as much variety as the set of all truths (i.e. propositions) that the agent can respond to.” (p. 107)
- **Generative ability.** “The situation here is not really strange. The underlying phenomenon is the generative ability of computational systems, which involves an active process working on an initially given data structure. Knowledge is the posited extensive form of all that can be obtained potentially from the process.” (p. 108)